

```
%% T5 - Sinais e Sistemas (Transformada de Laplace):  
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close all  
clear all  
clc
```

```
%% Atividade 1: Transformada de Laplace:
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```
syms t s
```

```
% Questão 1.1:  $g_a(t) = (1 + e^t)u(t)$ :  
disp('1. Transformada de Laplace de  $g_a(t)$ :');
```

1. Transformada de Laplace de $g_a(t)$:

```
ga = 1 + exp(t);  
Ga = laplace(ga);  
Ga
```

Ga =

$$\frac{1}{s-1} + \frac{1}{s}$$

```
% Questão 1.2:  $g_b(t) = (t \cdot e^{-3t} + e^{-5t})u(t)$ :  
disp('2. Transformada de Laplace de  $g_b(t)$ :');
```

2. Transformada de Laplace de $g_b(t)$:

```
gb = t*exp(-3*t) + exp(-5*t);  
Gb = laplace(gb);  
Gb
```

Gb =

$$\frac{1}{(s+3)^2} + \frac{1}{s+5}$$

```
% Questão 1.3:  $g_c(t) = e^{-t} \sin(3t)u(t)$ :  
disp('3. Transformada de Laplace de  $g_c(t)$ :');
```

3. Transformada de Laplace de $g_c(t)$:

```
gc = exp(-t)*sin(3*t);  
Gc = laplace(gc);  
Gc
```

Gc =

$$\frac{3}{(s+1)^2+9}$$

%% Atividade 2: Transformada Inversa de Laplace:

% Questão 2.1: $G_a(s) = 1 / (s+3)^2$:
disp('1. Transformada Inversa de $G_a(s)$:');

1. Transformada Inversa de $G_a(s)$:

```
Ga_s = 1 / (s+3)^2;  
ga_t = ilaplace(Ga_s);  
ga_t
```

$$ga_t = t e^{-3t}$$

% Questão 2.2: $G_b(s) = (s+1) / (s+7)$:
disp('2. Transformada Inversa de $G_b(s)$:');

2. Transformada Inversa de $G_b(s)$:

```
Gb_s = (s+1) / (s+7);  
gb_t = ilaplace(Gb_s);  
gb_t
```

$$gb_t = \delta(t) - 6e^{-7t}$$

% Questão 2.3: $G_c(s) = s / ((s+1)(s+2))$:
disp('3. Transformada Inversa de $G_c(s)$:');

3. Transformada Inversa de $G_c(s)$:

```
Gc_s = s / ((s+1)*(s+2));  
gc_t = ilaplace(Gc_s);  
gc_t
```

$$gc_t = 2e^{-2t} - e^{-t}$$